

A cubic lower bound for Electrostatic Equilibria: Failure of Maxwell's conjecture for infinitely many n

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Abstract

For distinct sites $a_1, \dots, a_n \in \mathbb{R}^3$ and nonzero real charges $q_1, \dots, q_n$, let $V_q(x) = \sum_{i=1}^n \frac{q_i}{|x - a_i|}$, and let $M(n)$ be the supremum of $\#\text{Crit}(V_q)$ over configurations with finite critical set. We prove

$$M(n) \geq \left(\frac{1}{48} - o(1) \right) n^3.$$

In particular, we get $M(n) > (n-1)^2$ for $n = 46$ and for every $n \geq 48$. Since all equilibria in our final configurations are nondegenerate, this disproves the standard Morse formulation of the bound, commonly known as Maxwell's conjecture.

Our construction starts with L charges on each of two perpendicular rays, where $n = 2L$. A charge on a ray contributes, in every degree, only the solid zonal harmonic about that ray. Alternating the two axes between consecutive degrees and separating the degrees onto different spatial scales makes the local model at scale k the sum of perpendicular solid zonal harmonics of degrees k and $k+1$. Its off-plane critical points arise from pairs (α, β) of zeros of P'_k and P'_{k+1} satisfying $\alpha^2 + \beta^2 < 1$ and a radial sign condition. The sign condition selects one member of each antipodal pair, while reflection across the coordinate plane produces two points. Consequently their total number is $g_k = (\frac{1}{2} + o(1))k^2$, and all these points are nondegenerate. A small generic perturbation then gives nonzero charges, a nonzero total charge, and a finite Morse critical set without losing any of the counted equilibria.

1 Introduction

1.1 The problem and its motivation

How complicated can the zero set of a field generated by finitely many Coulomb monopoles be? Let $a_1, \dots, a_n \in \mathbb{R}^3$ be distinct sites and let $q_1, \dots, q_n \in \mathbb{R} \setminus \{0\}$. On $\Omega = \mathbb{R}^3 \setminus \{a_1, \dots, a_n\}$, the potential and its equilibrium set are

$$V_q(x) = \sum_{i=1}^n \frac{q_i}{|x - a_i|}, \quad \text{Crit}(V_q) = \{x \in \Omega : \nabla V_q(x) = 0\}.$$

The electric field is $-\nabla V_q$, so $\text{Crit}(V_q)$ is precisely the set of positions at which a test charge experiences zero force. At the same time, V_q is harmonic on Ω . Hence $\text{tr Hess } V_q = 0$, and every nondegenerate equilibrium is necessarily a saddle, of Morse index 1 or 2. The counting

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problem therefore lies at a natural meeting point of potential theory, Morse theory, real algebraic geometry, and the geometry of harmonic functions. Even the qualitative issues are subtle: the field equations contain algebraic radicals, signed configurations can have nonisolated critical sets, and local critical points created near one scale must be protected from interactions with all other scales.

For positive charges, Morse theory gives at least $n - 1$ equilibria for a generic configuration; see, for example, [13, 4]. The opposite extremal question—how many equilibria can occur—is the subject of this paper. We use the finite-set extremal function

$$M(n) = \sup\{\#\text{Crit}(V_q) : a_i \text{ distinct, } q_i \neq 0, \text{Crit}(V_q) \text{ finite}\}.$$

This convention is slightly broader than the classical Morse formulation, because the critical points in the definition of $M(n)$ need not be nondegenerate. Our lower bound is nevertheless realized by configurations for which every equilibrium is nondegenerate.

1.2 Historical background and related work

Maxwell discussed electrostatic equilibrium counting in [12, §113]. The precise modern formulation usually called the *Maxwell conjecture*—that a potential with n point charges and only nondegenerate equilibria has at most $(n - 1)^2$ of them—was extracted from that discussion by Gabrielov, Novikov, and Shapiro [6]. Early Morse-theoretic work on nondegenerate Newtonian potentials includes Kiang [9]. Morse and Cairns later proved that, with the charge strengths and all but one site fixed, nondegeneracy holds for almost every position of the remaining site [13, Theorem 6.2]. A broader historical account of equilibrium questions in classical electrostatics is given in [1].

The first general finite upper bound for nondegenerate configurations was obtained by Gabrielov, Novikov, and Shapiro using Khovanskii’s fewnomial theory; their bound is independent of the ambient dimension, and they obtained the sharper bound 12 for three charges [6]. They also proposed a Voronoi-theoretic description for nonnegative charges. Zolotov subsequently used the Thom–Milnor theorem to obtain the exponential bound $5 \cdot 9^{n+3}$ [21]. Edelsbrunner, Fillmore, and Oliveira then gave the Bézout bound $2^n(3n - 2)^3$ under the generic-position hypothesis in their theorem, and disproved the proposed Voronoi upper bound [4].

Several sharper or more explicit results are known under planar, symmetric, or few-charge hypotheses. Killian obtained an improved upper bound for coplanar charges [10], and Peretz treated the three-charge problem by the argument principle [14]. Bilogliadov analyzed equal positive charges at the roots of unity for a family of Riesz potentials [3]. Tsai proved the proposed bound for equilateral and isosceles-right three-site geometries [17] and later gave a complete count for three charges of equal absolute magnitude [18]; Uteshev and Yashina developed exact algebraic and bifurcation methods for three and four planar charges [20]; and Erdélyi–Rosenblatt–Rosenblatt studied the structure and finiteness of field zero sets, including the case of collinear planar sources [5]. Lee and Tsai exhibited four coplanar charges with nine planar equilibria, attaining $(4 - 1)^2$ in that restricted setting [11]. On the algorithmic side, Ghosh, Goldberg, and Hollender recently initiated the complexity-theoretic study of computing approximate equilibria [7].

The lower-bound picture changed very recently. Edelsbrunner, Fillmore, and Oliveira constructed unit positive-charge configurations with $n_l = 8^l$ sites and exactly $\frac{25}{7}(n_l - 1)$ equilibria [4]. Arathoon, Ball, and Kvalheim then constructed five positive charges with at least 24 nondegenerate equilibria, thereby disproving the Maxwell bound; their iteration gives, for $n = 3 + 2m$,

at least $4 + 20m = 10n - 26$ equilibria [2]. These positive-charge lower bounds are linear in n . The result below is of a different nature: allowing charges of both signs produces cubic growth.

1.3 Main result

Let P_l denote the Legendre polynomial normalized by $P_l(1) = 1$, and, for a polynomial f , let $Z(f)$ be its set of zeros in $(-1, 1)$. For a unit vector w and $x \neq 0$, define

$$Z_l^w(x) = |x|^l P_l\left(\frac{\langle x, w \rangle}{|x|}\right).$$

This expression extends polynomially across $x = 0$ and is a harmonic homogeneous polynomial of degree l , called the solid zonal harmonic about w . Throughout, $u = e_3$ and $v = e_1$.

Theorem 1.1 (Main theorem). *Let $L \geq 2$ and set $n = 2L$. There are n distinct sites and nonzero real charges, with nonzero total charge and with every equilibrium nondegenerate, such that*

$$\# \text{Crit}(V_q) \geq S_L := \sum_{k=1}^{L-1} g_k, \quad g_k := \#\{(\alpha, \beta) \in Z(P'_k) \times Z(P'_{k+1}) : \alpha^2 + \beta^2 < 1\}.$$

Moreover,

$$g_k \geq \frac{k^2}{2} - 5k - 6 \quad (k \geq 3), \quad g_k = \left(\frac{1}{2} + o(1)\right) k^2.$$

Consequently, as $N \rightarrow \infty$ through all positive integers,

$$M(N) \geq \left(\frac{1}{48} - o(1)\right) N^3.$$

In particular, $M(46) > 45^2$ and $M(N) > (N - 1)^2$ for every $N \geq 48$.

Our construction does not settle $n = 47$. It is also important that the charges are allowed to have both signs; the theorem makes no corresponding cubic claim for positive charges.

1.4 Overview of the proof

Our proof builds many nondegenerate local equilibria and separates them so that they persist simultaneously. Its main steps are as follows.

1. *Moment control on two rays.* The multipole expansion of a charge at tw contains in degree l only the solid zonal harmonic about w , with coefficient $t^{-(l+1)}$. For L distinct sites on one ray, the map from the L charge strengths to the coefficients in degrees $1, \dots, L$ is a Vandermonde isomorphism. Thus two perpendicular rays allow us to prescribe one zonal coefficient about each of two axes in every relevant degree.
2. *A two-degree local model.* For perpendicular unit vectors u and v , consider $aZ_k^u + bZ_{k+1}^v$. Away from the coordinate plane, its angular critical-point equations separate as $P'_k(\alpha) = P'_{k+1}(\beta) = 0$. Such a pair gives two reflected critical points exactly when $\alpha^2 + \beta^2 < 1$ and a radial sign condition holds. Antipodal symmetry removes the sign condition from the final count, leaving exactly g_k points. Legendre's differential equation diagonalizes the Hessian at these points and proves that all of them are nondegenerate.

3. *Quadratic production at each rung.* After writing $\alpha = -\sin \varphi$ and $\beta = -\sin \psi$, the disk condition becomes the exact angular diamond $|\varphi| + |\psi| < \pi/2$. Standard localization of Legendre zeros turns the count into a lattice-point estimate in this diamond and gives $g_k = \frac{1}{2}k^2 + O(k)$. Summing over $k < L$ therefore yields $S_L = (\frac{1}{6} + o(1))L^3$.
4. *Scale separation and Coulomb realization.* We alternate the two axes in consecutive degrees and weight the degree- l term by $T^{-l(l-1)/2}$. On the spatial scale $|x| \asymp T^k$, the degrees k and $k+1$ dominate in C^2 , while every other degree is smaller by a power of T . The local equilibria from different rungs therefore persist in disjoint annuli. The ray moment isomorphisms realize this harmonic polynomial as the rescaled Taylor limit of an actual $2L$ -charge Coulomb potential. Interior estimates for harmonic functions control the full multipole tail in C^2 .
5. *Completion and finite thresholds.* Small perturbations make every charge and the total charge nonzero while preserving the marked equilibria. A transversality argument in site space makes all equilibria nondegenerate; the nonzero monopole term at infinity then makes the critical set finite. The same argument permits padding by additional sites and transfers the even construction to all integers. The asymptotic estimate gives the coefficient $1/48$, while exact rational root isolation for $2 \leq k \leq 26$ gives the stated crossover at 46 and at every $n \geq 48$.

1.5 Organization and notation

Section 2 proves the ray moment isomorphism. Section 3 classifies and counts the critical points of the two-term zonal model, including the asymptotic and exact finite estimates for g_k . Section 4 assembles these models on separated scales and realizes the resulting harmonic jet by point charges. Section 5 establishes finiteness, genericity, and padding and then proves the main theorem. Section 6 explains why alternating two axes is essential.

We use the standard facts that $|P_l| \leq 1$ on $[-1, 1]$, that $P_l(-t) = (-1)^l P_l(t)$, and that all zeros of P_l are simple and lie in $(-1, 1)$.

2 Ray moments

Lemma 2.1 (Rank-one excitation). *Let w be a unit vector and $t > 0$. For $|x| < t$,*

$$\frac{1}{|x - tw|} = \sum_{l=0}^{\infty} t^{-(l+1)} Z_l^w(x).$$

Consequently, if $q_1, \dots, q_L$ are placed at the distinct sites $t_1 w, \dots, t_L w$ with $t_i > 0$, then for $|x| < \min_i t_i$

$$\sum_{i=1}^L \frac{q_i}{|x - t_i w|} = \sum_{l=0}^{\infty} c_l Z_l^w(x), \quad c_l = \sum_{i=1}^L q_i t_i^{-(l+1)};$$

that is, the charges contribute $c_l Z_l^w$ in degree l . The map

$$(q_1, \dots, q_L) \longmapsto (c_1, \dots, c_L)$$

is a linear isomorphism of $\mathbb{R}^L$.

Proof. The series is the generating function for the Legendre polynomials, with $|x|/t$ as the radial variable; it converges absolutely and locally uniformly for $|x| < t$, which justifies summing the L expansions term by term. Put $s_i = t_i^{-1}$. The matrix of the moment map, with row index $1 \leq l \leq L$ and column index $1 \leq i \leq L$, is

$$[t_i^{-(l+1)}] * l, i = 1^L = [s_i^{l-1}] * l, i = 1^L \text{diag}(s_1^2, \dots, s_L^2).$$

The first factor is a Vandermonde matrix in the distinct numbers s_i , and the second is invertible. $\square$

Thus L charges on one ray control one scalar coefficient in each of degrees $1, \dots, L$. We use only this restricted moment map, not the ability to prescribe a general harmonic jet.

3 The two-term model

Lemma 3.1. *Let $k \geq 2$, let $a, b \in \mathbb{R} \setminus \{0\}$, and set*

$$m = aZ_k^u + bZ_{k+1}^v, \quad u = e_3, \quad v = e_1.$$

The critical points of m outside the plane $\{x_2 = 0\}$ are in bijection with

$$\left\{ (\alpha, \beta) \in Z(P'_k) \times Z(P'_{k+1}) : \begin{array}{l} \alpha^2 + \beta^2 < 1, \\ \frac{aP_k(\alpha)}{bP_{k+1}(\beta)} < 0 \end{array} \right\} \times \{\pm 1\}.$$

For such a pair, the two points are

$$x = r(\beta, \pm \sqrt{1 - \alpha^2 - \beta^2}, \alpha), \quad r = -\frac{k a P_k(\alpha)}{(k+1) b P_{k+1}(\beta)}.$$

They are nondegenerate and have Morse index 1 or 2. Their total number is g_k , independently of the signs of a and b .

Proof. On the half-space $\{x_2 > 0\}$, the map

$$(r, w_1, w_3) \mapsto r(w_1, \sqrt{1 - w_1^2 - w_3^2}, w_3)$$

is a diffeomorphism from $(0, \infty) \times D$ onto $\{x_2 > 0\}$, where $D = \{(w_1, w_3) : w_1^2 + w_3^2 < 1\}$. Note first that m depends on x only through x_1, x_3 and $|x|$, hence only through x_1, x_3 and x_2^2 ; so m is invariant under the reflection $x_2 \mapsto -x_2$, and it suffices to analyse $\{x_2 > 0\}$. In these coordinates,

$$m = ar^k P_k(w_3) + br^{k+1} P_{k+1}(w_1).$$

The two angular derivatives are $ar^k P'_k(w_3)$ and $br^{k+1} P'_{k+1}(w_1)$; since $r > 0$ and $a, b \neq 0$, they vanish precisely when $P'_k(w_3) = P'_{k+1}(w_1) = 0$. Write $(w_3, w_1) = (\alpha, \beta)$. Membership in the chart requires $(\beta, \alpha) \in D$, which is exactly the disc condition $\alpha^2 + \beta^2 < 1$. The zeros of P_l are simple, so P_l and P'_l have no common zero and $P_k(\alpha), P_{k+1}(\beta)$ are nonzero. The radial derivative is $kar^{k-1} P_k(\alpha) + (k+1)br^k P_{k+1}(\beta)$, which vanishes exactly at the value r displayed above; that value lies in the chart, that is $r > 0$, exactly under the stated sign condition. Reflection across $\{x_2 = 0\}$ gives the second point.

At such a critical point use the coordinate order (r, w_3, w_1) and put

$$A = aP_k(\alpha), \quad B = bP_{k+1}(\beta).$$

Every mixed second derivative carries a factor $P'_k(\alpha)$ or $P'_{k+1}(\beta)$ and hence vanishes, so the Hessian of m in these coordinates is diagonal; using the radial equation in the form $kA = -(k+1)rB$ for the first entry, it equals

$$\text{diag}\left(-kr^{k-2}A, r^k aP''_k(\alpha), r^{k+1} bP''_{k+1}(\beta)\right).$$

At a zero $t \in (-1, 1)$ of P'_l , Legendre's differential equation gives

$$(1-t^2)P''_l(t) = -l(l+1)P_l(t),$$

so that

$$aP''_k(\alpha) = -\frac{k(k+1)}{1-\alpha^2}A, \quad bP''_{k+1}(\beta) = -\frac{(k+1)(k+2)}{1-\beta^2}B.$$

Since $r > 0$, the relation $kA = -(k+1)rB$ forces $\text{sgn } B = -\text{sgn } A$. The three diagonal signs are therefore

$$(-\text{sgn } A, -\text{sgn } A, +\text{sgn } A),$$

and none is zero. A change of coordinates acts on the Hessian at a critical point by congruence, so by Sylvester's law of inertia this computation proves both nondegeneracy and the asserted Morse index in Cartesian coordinates.

It remains to count. Write

$$G_k = \{(\alpha, \beta) \in Z(P'_k) \times Z(P'_{k+1}) : \alpha^2 + \beta^2 < 1\},$$

so that $\#G_k = g_k$, and let $\sigma(\alpha, \beta) = (-\alpha, -\beta)$. Since $P_l(-t) = (-1)^l P_l(t)$, we have $P'_l(-t) = (-1)^{l+1} P'_l(t)$, so $Z(P'_k)$ and $Z(P'_{k+1})$ are symmetric about 0; as σ also preserves the disc condition, σ maps G_k to itself. Because P_k and P_{k+1} have opposite parity, the ratio $aP'_k(\alpha)/(bP'_{k+1}(\beta))$ changes sign under σ . That ratio is finite and nonzero, so it cannot equal its own negative; hence σ has no fixed point in G_k . (Equally directly: a fixed point would be $(0, 0)$, and $P'_l(0) = 0$ exactly when l is even, so P'_k and P'_{k+1} cannot both vanish at 0.) Therefore σ splits G_k into $g_k/2$ orbits of size two, and exactly one member of each orbit satisfies the sign condition. The reflection $x_2 \mapsto -x_2$ restores the factor 2, so the number of off-plane critical points is exactly g_k . $\square$

Lemma 3.2. *For every $k \geq 3$,*

$$g_k \geq \frac{k^2}{2} - 5k - 6,$$

and $g_k = (\frac{1}{2} + o(1))k^2$.

Proof. Put $m = k + \frac{1}{2}$ and $m' = k + \frac{3}{2}$. Write $\alpha = \cos \theta = -\sin \varphi$, where $\varphi = \theta - \pi/2 \in (-\pi/2, \pi/2)$, and similarly write $\beta = -\sin \psi$. Monotonicity of $\sin$ on $[0, \pi/2]$ gives the exact equivalence

$$\alpha^2 + \beta^2 < 1 \iff |\varphi| + |\psi| < \frac{\pi}{2}. \quad (1)$$

Thus the admissible angular region is the open diamond of area $\pi^2/2$.

Let $0 < \theta_1 < \dots < \theta_k < \pi$ be the zeros of $P_k(\cos \theta)$. A standard localization of the Legendre zeros [15, Theorem 6.21.2] states that

$$\left(j - \frac{1}{2}\right) \frac{\pi}{m} \leq \theta_j \leq j \frac{\pi}{m}, \quad 1 \leq j \leq k.$$

Rolle's theorem gives angles $\xi_j \in (\theta_j, \theta_{j+1})$ satisfying $P'_k(\cos \xi_j) = 0$, for $j = 1, \dots, k-1$; since $\deg P'_k = k-1$, these are all the zeros of P'_k in angular coordinates. The inclusion $\xi_j \in (\theta_j, \theta_{j+1})$ is strict, so

$$\left(j - \frac{1}{2}\right) \frac{\pi}{m} < \xi_j < (j+1) \frac{\pi}{m}$$

holds strictly even though the cited localization was stated with $\leq$. Since $\pi/2 = (\pi/m)(k/2 + 1/4)$, the angles $\varphi_j = \xi_j - \pi/2$ obey the *two-sided* estimate

$$\left|\varphi_j - \frac{\pi}{m} p_j\right| < \frac{3\pi}{4m}, \quad p_j := j - \frac{k}{2}, \quad 1 \leq j \leq k-1. \quad (2)$$

The zeros ψ_i of P'_{k+1} , written in the same way, obey

$$\left|\psi_i - \frac{\pi}{m'} q_i\right| < \frac{3\pi}{4m'}, \quad q_i := i - \frac{k+1}{2}, \quad 1 \leq i \leq k. \quad (3)$$

By (1) and the upper halves of (2)–(3), the pair (j, i) is admissible whenever

$$\frac{|p_j|}{m} + \frac{|q_i|}{m'} < \tau, \quad \tau = \frac{1}{2} - \frac{3}{4} \left(\frac{1}{m} + \frac{1}{m'} \right). \quad (4)$$

We now count this sufficient set. As j runs from 1 to $k-1$ the number $p = p_j$ runs through a step-1 lattice symmetric about 0 with $\max |p| = k/2 - 1$, and as i runs from 1 to k the number $q = q_i$ runs through a step-1 lattice symmetric about 0 with $\max |q| = (k-1)/2$. These index ranges impose no further restriction: every lattice pair (p, q) satisfying (4) does come from indices $1 \leq j \leq k-1$ and $1 \leq i \leq k$, since (4) forces $|p| < m\tau$ and $|q| < m'\tau$, while

$$m\tau = \frac{k}{2} - \frac{1}{2} - \frac{3m}{4m'} < \frac{k}{2} - 1, \quad m'\tau = \frac{k}{2} - \frac{3m'}{4m} < \frac{k-1}{2};$$

here both inequalities use $\frac{2}{3} < m/m' < 1$, which holds for every $k \geq 2$. For fixed p , the admissible q lie in an interval of length $2m'(\tau - |p|/m)$. An open interval of length ℓ contains at least $\ell - 1$ points of any step-1 lattice. Restrict to $|p| \leq R := m\tau - 1$, which is nonnegative for $k \geq 5$. There are at least $2R - 1$ such p 's. Also $m'\tau > m\tau \geq 1$, so $2m'\tau - 1 > 0$; this positivity is what lets the (possibly negative) bound $2R - 1$ be substituted for the number of such p 's below. Moreover,

$$\sum_{|p| \leq R} |p| \leq R^2 + R + \frac{1}{4}. \quad (5)$$

Indeed, for the integer lattice the left side is $h(h+1)$, where $h = \lfloor R \rfloor$; for the half-integer lattice it is $(h+1)^2$, where $h+1/2$ is the largest nonnegative lattice point not exceeding R . If the half-integer set is empty, the left side is instead 0. All three cases give (5).

Let $N(k)$ denote the number of lattice pairs satisfying (4). Summing the interval bound over the values $|p| \leq R$ alone, and discarding the remaining p 's, gives

$$\begin{aligned} N(k) &\geq (2R-1)(2m'\tau-1) - \frac{2m'}{m} \left(R^2 + R + \frac{1}{4} \right) \\ &= 2mm'\tau^2 - 2m\tau - 4m'\tau + 3 - \frac{m'}{2m}. \end{aligned}$$

For $k \geq 5$, both τ and $\frac{1}{2} - \frac{3}{2m}$ are positive. Since $\tau > \frac{1}{2} - \frac{3}{2m}$ and $m'/(2m) < 1$, we obtain

$$2mm'\tau^2 \geq \frac{mm'}{2} - 3m', \quad 2m\tau + 4m'\tau < m + 2m' = 3k + \frac{7}{2}.$$

Substituting $\frac{mm'}{2} = \frac{k^2}{2} + k + \frac{3}{8}$ and $3m' = 3k + \frac{9}{2}$ therefore gives

$$N(k) > \left(\frac{k^2}{2} + k + \frac{3}{8} \right) - \left(3k + \frac{9}{2} \right) - \left(3k + \frac{7}{2} \right) + 3 - 1 = \frac{k^2}{2} - 5k - \frac{45}{8}.$$

Since (4) is sufficient for admissibility and distinct lattice pairs come from distinct index pairs, $g_k \geq N(k)$, so

$$g_k > \frac{k^2}{2} - 5k - \frac{45}{8} > \frac{k^2}{2} - 5k - 6 \quad (k \geq 5).$$

For $k = 3, 4$ the asserted lower bound is negative and hence automatic.

For the asymptotic a matching upper bound is needed. By (1) and the *lower* halves of (2)–(3), every admissible pair satisfies the necessary condition

$$\frac{|p_j|}{m} + \frac{|q_i|}{m'} < \tau^+, \quad \tau^+ = \frac{1}{2} + \frac{3}{4} \left(\frac{1}{m} + \frac{1}{m'} \right). \quad (6)$$

Distinct index pairs give distinct lattice pairs, so g_k is at most the number of lattice pairs (p, q) obeying (6). Assign to each such pair the closed unit square centred at it. Both lattices have step 1, so these squares have pairwise disjoint interiors, and each is contained in the diamond

$$\Delta = \left\{ (\eta, \zeta) : \frac{|\eta|}{m} + \frac{|\zeta|}{m'} \leq \tau^{++} \right\}, \quad \tau^{++} = \tau^+ + \frac{1}{2m} + \frac{1}{2m'} = \frac{1}{2} + \frac{5}{4} \left(\frac{1}{m} + \frac{1}{m'} \right),$$

whose area is $2mm'(\tau^{++})^2$. Comparing total areas,

$$g_k \leq 2mm'(\tau^{++})^2 = \frac{mm'}{2} + \frac{5}{2}(m+m') + \frac{25}{8} \cdot \frac{(m+m')^2}{mm'}.$$

Here $mm' = k^2 + 2k + \frac{3}{4}$ and $m + m' = 2k + 2$, so the first two terms are $\frac{1}{2}k^2 + 6k + \frac{43}{8}$, while the third equals $\frac{25}{8} \cdot 4(k+1)^2 / ((k+1)^2 - \frac{1}{4})$, which decreases in k and is therefore at most its value $\frac{25}{8} \cdot \frac{144}{35} = \frac{90}{7} < 13$ at $k = 2$. Hence

$$\frac{k^2}{2} - 5k - 6 \leq g_k \leq \frac{k^2}{2} + 6k + 19 \quad (k \geq 3),$$

so $g_k = \frac{1}{2}k^2 + O(k)$; in particular $g_k = (\frac{1}{2} + o(1))k^2$, as claimed. The constant $\frac{1}{2}$ is the ratio of the area of the diamond (1) to that of the square $(-\pi/2, \pi/2)^2$ swept out by the two angular meshes. $\square$

Lemma 3.3 (Exact finite counts). *The values needed for the finite thresholds in Theorem 1.1 are*

$$(g_2, \dots, g_{14}) = (2, 6, 8, 16, 22, 30, 36, 48, 58, 70, 80, 96, 110),$$

$$(g_{15}, \dots, g_{26}) = (126, 140, 160, 178, 198, 216, 240, 262, 286, 308, 336, 362).$$

In particular,

$$S_{23} = 2102, \quad S_{24} = 2388.$$

Proof. This is a finite exact computation. For each $2 \leq k \leq 26$, exact real-root isolation gives pairwise disjoint rational intervals containing the simple zeros of P'_k and P'_{k+1} . Rational interval arithmetic then decides whether $\alpha^2 + \beta^2 < 1$ for every pair; refining the isolating intervals decides every pair, because no pair satisfies $\alpha^2 + \beta^2 = 1$. That last fact is itself certified exactly, and independently of any tolerance, by the nonvanishing of the iterated resultant

$$\text{Res} * \beta(P * k + 1'(\beta), \text{Res}_\alpha(P'_k(\alpha), \alpha^2 + \beta^2 - 1)).$$

The ancillary file `verify_counts.py` implements these steps using only rational polynomial arithmetic, verifies all root multiplicities and inequalities, and prints the values above and their partial sums. It has been tested with SymPy 1.14.0 [16]; the accompanying JSON certificate records the version used and all isolating intervals and decisions. Thus this lemma does not rely on floating-point root finding. $\square$

4 The scale-separated chain

Lemma 4.1 (Scale separation). *Let ζ_l be harmonic and homogeneous of degree l , for $1 \leq l \leq L$, and let*

$$P_T = \sum_{l=1}^L T^{-l(l-1)/2} \zeta_l, \quad T > 1.$$

Suppose that, for each $1 \leq k < L$, the polynomial $\zeta_k + \zeta_{k+1}$ has a specified set of N_k nondegenerate critical points in the annulus $W^{-1} \leq |Y| \leq W$, where $W \geq 1$. For all sufficiently large T , the polynomial P_T has at least $\sum_{k=1}^{L-1} N_k$ nondegenerate critical points.

Proof. Fix k , substitute $x = T^k Y$, and divide by $T^{k(k+1)/2}$. Homogeneity gives

$$T^{-k(k+1)/2} P_T(T^k Y) = \zeta_k(Y) + \zeta_{k+1}(Y) + \sum_{l \neq k, k+1} T^{-(l-k)(l-k-1)/2} \zeta_l(Y).$$

For $l \neq k, k+1$, the exponent $(l-k)(l-k-1)/2$ is at least 1. Therefore the final sum tends to zero in C^2 on the compact annulus $(2W)^{-1} \leq |Y| \leq 2W$. Nondegenerate zeros of the gradient persist under C^1 -small perturbations, so for T large each of the N_k selected points has a nearby nondegenerate critical point of the left-hand side; these stay pairwise distinct and, having moved by an arbitrarily small amount from $W^{-1} \leq |Y| \leq W$, remain in $(2W)^{-1} \leq |Y| \leq 2W$. In the original variable they lie in

$$\frac{T^k}{2W} \leq |x| \leq 2WT^k.$$

Since $Y \mapsto T^k Y$ scales the left-hand side by a positive constant, these are nondegenerate critical points of P_T itself. The finitely many choices of k admit one common lower bound on T , and if $T > (2W)^2$ these annuli are pairwise disjoint. Hence no point is counted at two scales. $\square$

Apply the lemma to

$$\zeta_l = \begin{cases} Z_l^u, & l \text{ even}, \\ Z_l^v, & l \text{ odd}. \end{cases} \quad (7)$$

For $k \geq 2$ even, $\zeta_k + \zeta_{k+1} = Z_k^u + Z_{k+1}^v$ and Lemma 3.1 with $a = b = 1$ supplies $N_k = g_k$ nondegenerate points. For $k \geq 3$ odd, $\zeta_k + \zeta_{k+1} = Z_k^v + Z_{k+1}^u$; composing with the orthogonal involution σ that exchanges e_1 and e_3 turns this into $Z_k^u + Z_{k+1}^v$, since $Z_l^w \circ \sigma = Z_l^{\sigma w}$. As σ preserves $|Y|$, nondegeneracy and the number of critical points, Lemma 3.1 again supplies $N_k = g_k$ points. We take $N_1 = g_1 = 0$, which is consistent because P_1' has no zeros. For fixed L these are finitely many points at positive radii, so a common W exists. Lemma 4.1 consequently gives

$$\# \text{Crit}(P_T) \geq S_L \quad (8)$$

for all sufficiently large T .

Lemma 4.2 (Coulomb realization). *Fix L and then choose T so that (8) holds. Fix distinct $t_1, \dots, t_L \in [1, 2]$ and distinct $s_1, \dots, s_L \in [1, 2]$. There is a family $(V_\varepsilon)_{0 < \varepsilon \leq 1}$ of Coulomb potentials, all with the same $2L$ distinct sites $t_i u$ and $s_i v$, such that the rescalings $Y \mapsto V_\varepsilon(\varepsilon Y) - V_\varepsilon(0)$ converge to P_T in C^2 on a ball containing all the critical points counted in (8). Hence, for all sufficiently small positive ε , the potential V_ε has at least S_L nondegenerate critical points.*

Proof. For $1 \leq l \leq L$, prescribe the ray moments

$$c_l = \begin{cases} \varepsilon^{-l} T^{-l(l-1)/2}, & l \text{ even}, \\ 0, & l \text{ odd}, \end{cases} \quad d_l = \begin{cases} 0, & l \text{ even}, \\ \varepsilon^{-l} T^{-l(l-1)/2}, & l \text{ odd}. \end{cases}$$

Lemma 2.1 gives charge vectors $q(\varepsilon) = (q_1, \dots, q_L)$ on the u -ray and $p(\varepsilon) = (p_1, \dots, p_L)$ on the v -ray. Let

$$V_\varepsilon(x) = \sum_{i=1}^L \frac{q_i(\varepsilon)}{|x - t_i u|} + \sum_{i=1}^L \frac{p_i(\varepsilon)}{|x - s_i v|}.$$

Put $\rho = \min_i(t_i, s_i)$, so that $\rho \geq 1$; the ball $\{|x| < \rho\}$ contains no site, and the Taylor series from Lemma 2.1 converges there. Homogeneity and the prescribed moments give

$$V_\varepsilon(\varepsilon Y) - V_\varepsilon(0) = P_T(Y) + R_\varepsilon(Y), \quad R_\varepsilon(Y) = \sum_{l>L} \varepsilon^l (\tilde{c}_l Z_l^u(Y) + \tilde{d}_l Z_l^v(Y)),$$

where $\tilde{c}_l, \tilde{d}_l$ are the actual ray moments in degrees $l > L$.

We prove the required C^2 convergence. The two inverse moment maps are fixed after the sites are chosen, and $T^{-l(l-1)/2} \leq 1$ because $T > 1$. Hence, for $0 < \varepsilon \leq 1$,

$$\|q(\varepsilon)\|_1 + \|p(\varepsilon)\| * 1 \leq K \varepsilon^{-L}$$

for a constant K depending on L and the chosen sites. Since $|\tilde{c}_l| \leq \|q(\varepsilon)\|_1 \rho^{-(l+1)}$ and likewise for $\tilde{d}_l$,

$$|\tilde{c}_l| + |\tilde{d}_l| \leq K \varepsilon^{-L} \rho^{-(l+1)}.$$

Since $|Z_l^w(Y)| \leq |Y|^l$, the geometric tail $\sum_{l>L} (\varepsilon|Y|/\rho)^l \leq 2(\varepsilon|Y|/\rho)^{L+1}$ shows that, whenever $\varepsilon|Y| \leq \rho/2$,

$$|R * \varepsilon(Y)| \leq C \varepsilon^{-L} \rho^{-1} \left(\frac{\varepsilon|Y|}{\rho} \right)^{L+1}, \quad C = 2K.$$

Choose $R_0 \geq 1$ so that the ball B_{R_0} contains all the critical points of P_T selected in Lemma 4.1. For sufficiently small ε , the preceding estimate applies on B_{2R_0} and gives $\|R_\varepsilon\|_{C^0(B_{2R_0})} = O(\varepsilon)$. The remainder is harmonic there, so the standard interior derivative estimates for harmonic functions imply

$$\|R_\varepsilon\| * C^2(B * R_0) = O(\varepsilon).$$

Every selected critical point of P_T is nondegenerate; therefore all of them persist as distinct nondegenerate critical points of $P_T + R_\varepsilon$. Rescaling $x = \varepsilon Y$ preserves critical points and nondegeneracy, and it places them in $\{|x| \leq \varepsilon R_0\} \subset \{|x| < \rho\}$, which contains no site. This proves the lemma. $\square$

5 Finiteness, genericity, and padding

The two-ray potential constructed above already has the required marked nondegenerate equilibria. It may, however, have other degenerate equilibria, zero entries in its charge vector, or zero total charge. The next two lemmas remove all three possibilities without losing any marked point.

Lemma 5.1 (Finiteness). *Let V_q be generated by finitely many distinct sites and nonzero charges. If the total charge $Q = \sum_i q_i$ is nonzero and every critical point of V_q is nondegenerate, then $\text{Crit}(V_q)$ is finite.*

Proof. Fix a site a_i and choose a ball about it containing no other site. On this ball the field due to all other charges is bounded, whereas

$$\left| \nabla \frac{q_i}{|x - a_i|} \right| = \frac{|q_i|}{|x - a_i|^2}.$$

Thus a sufficiently small punctured neighborhood of a_i contains no critical point.

At infinity, the multipole expansion gives

$$\nabla V_q(x) = -\frac{Qx}{|x|^3} + O(|x|^{-3}).$$

Since $Q \neq 0$, the field is nonzero outside a sufficiently large ball. Consequently $\text{Crit}(V_q)$ is contained in a compact subset C of Ω . If $\text{Crit}(V_q)$ were infinite it would have an accumulation point $x_* \in C$, and $\nabla V_q(x_*) = 0$ by continuity, so x_* would be a nonisolated critical point. This contradicts the fact that nondegenerate critical points are isolated. $\square$

Lemma 5.2 (Morse completion and padding). *Suppose a Coulomb potential with n distinct sites and arbitrary real charge strengths has a finite set of K specified nondegenerate critical points. For every $m \geq n$, there is a configuration of m distinct sites and nonzero real charges, with nonzero total charge and with every critical point nondegenerate, that has at least K critical points.*

Proof. First choose pairwise disjoint persistence neighborhoods of the K marked points. Choose $m - n$ additional sites outside their closures, all distinct from the original sites and from one another. Give the new sites sufficiently small nonzero strengths, and perturb the original strengths by arbitrarily small amounts. The resulting change of potential is C^2 -small on the persistence

neighborhoods, because the new sites lie at positive distance from them; since the K marked critical points are nondegenerate, they persist, one in each neighborhood. The strengths can be chosen so that every one is nonzero and their sum Q is nonzero.

Fix this charge vector and vary the sites. Let $\mathcal{A}$ be the open configuration space of ordered, pairwise distinct sites, and define

$$\mathcal{E} = \{(a, x) : a \in \mathcal{A}, x \neq a_i \text{ for every } i\}, \quad F(a, x) = \nabla_x \sum_{i=1}^m \frac{q_i}{|x - a_i|}.$$

For each i ,

$$D_{a_i} F(a, x) = -q_i \text{Hess}_x \left(\frac{1}{|x - a_i|} \right).$$

The Hessian of $1/|x - a_i|$ has eigenvalues $2|x - a_i|^{-3}$, $-|x - a_i|^{-3}$, $-|x - a_i|^{-3}$, so, q_i being nonzero, $D_{a_i} F$ is invertible. Thus F is a submersion, so $\mathcal{Z} = F^{-1}(0)$ is a smooth submanifold of $\mathcal{E}$ of dimension $\dim \mathcal{E} - 3 = 3m$. Let $\pi : \mathcal{Z} \rightarrow \mathcal{A}$ be the projection. At $(a, x) \in \mathcal{Z}$, the derivative $D\pi(a, x)$ is surjective exactly when $D_x F(a, x)$ is invertible. Indeed, surjectivity of $D\pi$ says that for every $\dot{a}$ there is a $\dot{x}$ with

$$D_a F(a, x)\dot{a} + D_x F(a, x)\dot{x} = 0.$$

If $D_x F$ is invertible this is immediate. Conversely, since $D_a F$ is surjective, every $y \in \mathbb{R}^3$ has the form $D_a F\dot{a}$; surjectivity of $D\pi$ then gives $y = -D_x F\dot{x}$, so $D_x F$ is surjective and hence invertible. Writing $V_{q,a}$ for the potential with charge vector q and site vector a , we have $\pi^{-1}(a) = \{a\} \times \text{Crit}(V_{q,a})$, so this says that a is a regular value of π exactly when every equilibrium for that site configuration is nondegenerate. The map π is smooth between manifolds of the same dimension $3m$, so by Sard's theorem [8, Ch. 3] its critical values form a set of measure zero; hence the regular values of π are dense in the open set $\mathcal{A} \subset \mathbb{R}^{3m}$.

Choose such a site configuration arbitrarily close to the one already constructed. The sites remain distinct and, the potential depending smoothly on the sites in C^2 away from them, the K marked equilibria persist. The charge vector, and hence $Q \neq 0$, is unchanged. The critical set is finite by Lemma 5.1. $\square$

Proof of Theorem 1.1. For fixed L , Lemmas 4.1 and 4.2 produce a Coulomb potential with $2L$ sites and at least S_L marked nondegenerate critical points. Apply Lemma 5.2 with $m = 2L$. This gives the configuration asserted in the first part of the theorem.

By Lemma 3.2,

$$S_L = \sum_{k=1}^{L-1} g_k = \left(\frac{1}{6} + o(1) \right) L^3.$$

Since $n = 2L$, this gives $M(n) \geq (1/48 - o(1))n^3$ along the even integers. For $N = 2L + 1$, Lemma 5.2 pads the $2L$ -site construction to N sites without losing any marked equilibrium, so $M(N) \geq S_L$, and $S_L \sim L^3/6 \sim N^3/48$. This proves the asymptotic statement along all integers.

For the explicit thresholds, note first that $S_{L+1} - S_L = g_L$.

Even N . By Lemma 3.3, $S_{23} = 2102 > 2025 = 45^2$, so $M(46) \geq S_{23} > 45^2$. Passing from L to $L + 1$ raises the benchmark by

$$(2L + 1)^2 - (2L - 1)^2 = 8L,$$

so $S_L > (2L - 1)^2$ propagates to $S_{L+1} > (2L + 1)^2$ as soon as $g_L > 8L$. The exact table gives $g_{23}, g_{24}, g_{25}, g_{26} = 286, 308, 336, 362$, each exceeding the corresponding $8L = 184, 192, 200, 208$; and for $L \geq 27$ Lemma 3.2 gives $g_L \geq \frac{1}{2}L^2 - 5L - 6 > 8L$, the last inequality being $L^2 - 26L - 12 > 0$. By induction $S_L > (2L - 1)^2$ for every $L \geq 23$, that is, $M(N) > (N - 1)^2$ for every even $N \geq 46$.

Odd N . Here $N = 2L + 1$ and the benchmark is $(2L)^2$. By Lemma 3.3, $S_{24} = 2388 > 2304 = 48^2$, so $M(49) > 48^2$. Passing from L to $L + 1$ raises the benchmark by

$$(2L + 2)^2 - (2L)^2 = 8L + 4,$$

so the induction propagates as soon as $g_L > 8L + 4$. Again $g_{24}, g_{25}, g_{26} = 308, 336, 362$ exceed $8L + 4 = 196, 204, 212$, and for $L \geq 27$ Lemma 3.2 gives $g_L \geq \frac{1}{2}L^2 - 5L - 6 > 8L + 4$, the last inequality being $L^2 - 26L - 20 > 0$. Since Lemma 5.2 pads each $2L$ -site construction by one site while preserving its S_L marked equilibria, $M(2L + 1) \geq S_L > (2L)^2$ for every $L \geq 24$, that is, $M(N) > (N - 1)^2$ for every odd $N \geq 49$.

The two cases together give $M(46) > 45^2$ and $M(N) > (N - 1)^2$ for every $N \geq 48$, which is the final assertion. (They leave $N = 47$ open, since $S_{23} = 2102 < 2116 = 46^2$.) $\square$

6 Why two different axes are essential

The alternating axes are necessary. Consecutive zonal blocks about one axis cannot produce an off-axis critical point.

Proposition 6.1. *Let $a, b \neq 0$ and $k \geq 1$. The polynomial $aZ_k^u + bZ_{k+1}^u$ has no critical point outside the axis $\mathbb{R}u$.*

Proof. Use spherical coordinates (r, θ, ϕ) about u , with $\mu = \cos \theta$. Each Z_l^u is independent of ϕ , so in the orthonormal spherical frame (e_r, e_θ, e_ϕ) its e_ϕ -component vanishes and

$$\nabla Z_l^u = r^{l-1} (lP_l(\mu)e_r - \sin \theta P_l'(\mu)e_\theta).$$

For $0 < \theta < \pi$, the determinant of the two nonzero components in degrees k and $k + 1$ is $r^{2k-1} \sin \theta G_k(\mu)$, where

$$G_k = (k + 1)P_{k+1}P_k' - kP_kP_{k+1}'.$$

The identities $(1 - \mu^2)P_l'(\mu) = l(P_{l-1}(\mu) - \mu P_l(\mu))$, applied with $l = k$ and $l = k + 1$, give

$$(1 - \mu^2)G_k(\mu) = -k(k + 1)(P_k(\mu)^2 - P_{k-1}(\mu)P_{k+1}(\mu)).$$

Turán's inequality is strict for $-1 < \mu < 1$ [19], so these two gradients are linearly independent away from the axis. No nontrivial linear combination of them can vanish there. $\square$

In particular, a chain made entirely from coaxial consecutive zonal blocks has no off-axis contribution at any rung. Also, invariance under a rotation of order N about a fixed axis forces a spherical harmonic of degree l to have azimuthal orders divisible by N ; if $l < N$ the only such order is 0, so the degree- l block is zonal about that axis. This observation explains the obstruction for the low-degree rungs of a fixed C_N -symmetric construction. It does not assert that all higher-degree C_N -invariant blocks are zonal.

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